



MATH155: Sequences and Series

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Learning Objectives

- Find the n th term of an arithmetic sequence
- Find the n th term of a geometric sequence
- Identify the common difference or common ratio of a sequence

Simplify each expression completely. Show all steps and circle your final answer.

Arithmetic sequences

1. An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 2$. Find the 9th term.

$$a_1 = 4, d = 2, n = 9$$

Answer: _____

2. Find the common difference d of the arithmetic sequence: 2, 9, 16, ...

$$2, 9, 16, \dots$$

Answer: _____

3. An arithmetic sequence has first term $a_1 = 3$ and common difference $d = 5$. Find the 5th term.

$$a_1 = 3, d = 5, n = 5$$

Answer: _____

4. Find the common difference d of the arithmetic sequence: 7, 11, 15, ...

$$7, 11, 15, \dots$$

Answer: _____

5. An arithmetic sequence has first term $a_1 = 16$ and common difference $d = 8$. Find the 9th term.

$$a_1 = 16, d = 8, n = 9$$

Answer: _____

6. Find the common difference d of the arithmetic sequence: 5, 10, 15, ...

$$5, 10, 15, \dots$$

Answer: _____

7. An arithmetic sequence has first term $a_1 = 5$ and common difference $d = 5$. Find the 13th term.

$$a_1 = 5, d = 5, n = 13$$

Answer: _____

8. Find the common difference d of the arithmetic sequence: 3, 9, 15, ...

$$3, 9, 15, \dots$$

Answer: _____

9. An arithmetic sequence has first term $a_1 = 19$ and common difference $d = 3$. Find the 8th term.

$$a_1 = 19, d = 3, n = 8$$

Answer: _____

10. Find the common difference d of the arithmetic sequence: 5, 9, 13, ...

$$5, 9, 13, \dots$$

Answer: _____

11. An arithmetic sequence has first term $a_1 = 19$ and common difference $d = 3$. Find the 6th term.

$$a_1 = 19, d = 3, n = 6$$

Answer: _____

12. Find the common difference d of the arithmetic sequence: 9, 19, 29, ...

$$9, 19, 29, \dots$$

Answer: _____

13. An arithmetic sequence has first term $a_1 = 3$ and common difference $d = 4$. Find the 11th term.

$$a_1 = 3, d = 4, n = 11$$

Answer: _____

14. Find the common difference d of the arithmetic sequence: 3, 7, 11, ...

$$3, 7, 11, \dots$$

Answer: _____

15. An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 4$. Find the 12th term.

$$a_1 = 4, d = 4, n = 12$$

Answer: _____

16. Find the common difference d of the arithmetic sequence: 7, 10, 13, ...

7, 10, 13, ...

Answer: _____

Geometric sequences

17. A geometric sequence has first term $a_1 = 4$ and common ratio $r = 4$. Find the 7th term.

$a_1 = 4, r = 4, n = 7$

Answer: _____

18. Find the common ratio r of the geometric sequence: 3, 15, 75, ...

3, 15, 75, ...

Answer: _____

19. A geometric sequence has first term $a_1 = 5$ and common ratio $r = 3$. Find the 4th term.

$a_1 = 5, r = 3, n = 4$

Answer: _____

20. Find the common ratio r of the geometric sequence: 1, 4, 16, ...

1, 4, 16, ...

Answer: _____

21. A geometric sequence has first term $a_1 = 3$ and common ratio $r = 2$. Find the 7th term.

$a_1 = 3, r = 2, n = 7$

Answer: _____

22. Find the common ratio r of the geometric sequence: 2, 10, 50, ...

2, 10, 50, ...

Answer: _____

23. A geometric sequence has first term $a_1 = 5$ and common ratio $r = 2$. Find the 6th term.

$a_1 = 5, r = 2, n = 6$

Answer: _____

24. Find the common ratio r of the geometric sequence: 1, 4, 16, ...

1, 4, 16, ...

Answer: _____

25. A geometric sequence has first term $a_1 = 3$ and common ratio $r = 3$. Find the 7th term.

$$a_1 = 3, r = 3, n = 7$$

Answer: _____

26. Find the common ratio r of the geometric sequence: 4, 12, 36, ...

$$4, 12, 36, \dots$$

Answer: _____

27. A geometric sequence has first term $a_1 = 4$ and common ratio $r = 2$. Find the 7th term.

$$a_1 = 4, r = 2, n = 7$$

Answer: _____

28. Find the common ratio r of the geometric sequence: 4, 16, 64, ...

$$4, 16, 64, \dots$$

Answer: _____

29. A geometric sequence has first term $a_1 = 4$ and common ratio $r = 4$. Find the 7th term.

$$a_1 = 4, r = 4, n = 7$$

Answer: _____

30. Find the common ratio r of the geometric sequence: 1, 2, 4, ...

$$1, 2, 4, \dots$$

Answer: _____



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ANSWER KEY & SOLUTIONS

Topics: Arithmetic sequences, Geometric sequences. All answers verified by independent computation.

Solutions

Arithmetic sequences

1. An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 2$. Find the 9th term.

$$a_1 = 4, d = 2, n = 9$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_9 = 4 + (9 - 1)(2).$$

$$\rightarrow = 4 + 8 * 2 = 4 + 16 = 20.$$

Answer: $a_9 = 20$

2. Find the common difference d of the arithmetic sequence: 2, 9, 16, ...

$$2, 9, 16, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 9 - 2 = 7$.

→ Verify: $9 - 2 = 16 - 9 = 7$. The sequence is arithmetic.

Answer: $d = 7$

3. An arithmetic sequence has first term $a_1 = 3$ and common difference $d = 5$. Find the 5th term.

$$a_1 = 3, d = 5, n = 5$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_5 = 3 + (5 - 1)(5).$$

$$\rightarrow = 3 + 4 * 5 = 3 + 20 = 23.$$

Answer: $a_5 = 23$

4. Find the common difference d of the arithmetic sequence: 7, 11, 15, ...

$$7, 11, 15, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 11 - 7 = 4$.

→ Verify: $11 - 7 = 15 - 11 = 4$. The sequence is arithmetic.

Answer: $d = 4$

5. An arithmetic sequence has first term $a_1 = 16$ and common difference $d = 8$. Find the 9th term.

$$a_1 = 16, d = 8, n = 9$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_9 = 16 + (9 - 1)(8).$$

$$\rightarrow = 16 + 8 * 8 = 16 + 64 = 80.$$

Answer: $a_9 = 80$

6. Find the common difference d of the arithmetic sequence: 5, 10, 15, ...

$$5, 10, 15, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 10 - 5 = 5$.

→ Verify: $10 - 5 = 15 - 10 = 5$. The sequence is arithmetic.

Answer: $d = 5$

7. An arithmetic sequence has first term $a_1 = 5$ and common difference $d = 5$. Find the 13th term.

$$a_1 = 5, d = 5, n = 13$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_{13} = 5 + (13 - 1)(5).$$

$$\rightarrow = 5 + 12 * 5 = 5 + 60 = 65.$$

Answer: $a_{13} = 65$

8. Find the common difference d of the arithmetic sequence: 3, 9, 15, ...

$$3, 9, 15, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 9 - 3 = 6$.

→ Verify: $9 - 3 = 15 - 9 = 6$. The sequence is arithmetic.

Answer: $d = 6$

9. An arithmetic sequence has first term $a_1 = 19$ and common difference $d = 3$. Find the 8th term.

$$a_1 = 19, d = 3, n = 8$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_8 = 19 + (8 - 1)(3).$$

$$\rightarrow = 19 + 7 * 3 = 19 + 21 = 40.$$

Answer: $a_8 = 40$

10. Find the common difference d of the arithmetic sequence: 5, 9, 13, ...

$$5, 9, 13, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 9 - 5 = 4$.

→ Verify: $9 - 5 = 13 - 9 = 4$. The sequence is arithmetic.

Answer: $d = 4$

11. An arithmetic sequence has first term $a_1 = 19$ and common difference $d = 3$. Find the 6th term.

$$a_1 = 19, d = 3, n = 6$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_6 = 19 + (6 - 1)(3).$$

$$\rightarrow = 19 + 5 * 3 = 19 + 15 = 34.$$

Answer: $a_6 = 34$

12. Find the common difference d of the arithmetic sequence: 9, 19, 29, ...

$$9, 19, 29, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 19 - 9 = 10$.

→ Verify: $19 - 9 = 29 - 19 = 10$. The sequence is arithmetic.

Answer: $d = 10$

13. An arithmetic sequence has first term $a_1 = 3$ and common difference $d = 4$. Find the 11th term.

$$a_1 = 3, d = 4, n = 11$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_{11} = 3 + (11 - 1)(4).$$

$$\rightarrow = 3 + 10 * 4 = 3 + 40 = 43.$$

Answer: $a_{11} = 43$

14. Find the common difference d of the arithmetic sequence: 3, 7, 11, ...

$$3, 7, 11, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 7 - 3 = 4$.

→ Verify: $7 - 3 = 11 - 7 = 4$. The sequence is arithmetic.

Answer: $d = 4$

15. An arithmetic sequence has first term $a_1 = 4$ and common difference $d = 4$. Find the 12th term.

$$a_1 = 4, d = 4, n = 12$$

→ Use the formula $a_n = a_1 + (n - 1)d$.

$$\rightarrow a_{12} = 4 + (12 - 1)(4).$$

$$\rightarrow = 4 + 11 * 4 = 4 + 44 = 48.$$

Answer: $a_{12} = 48$

16. Find the common difference d of the arithmetic sequence: 7, 10, 13, ...

$$7, 10, 13, \dots$$

→ Subtract consecutive terms: $d = a_2 - a_1 = 10 - 7 = 3$.

→ Verify: $10 - 7 = 13 - 10 = 3$. The sequence is arithmetic.

Answer: $d = 3$

Geometric sequences

17. A geometric sequence has first term $a_1 = 4$ and common ratio $r = 4$. Find the 7th term.

$$a_1 = 4, r = 4, n = 7$$

→ Use the formula $a_n = a_1 * r^{(n - 1)}$.

$$\rightarrow a_7 = 4 * 4^{(7 - 1)} = 4 * 4^6.$$

$$\rightarrow = 4 * 4096 = 16384.$$

Answer: $a_7 = 16384$

18. Find the common ratio r of the geometric sequence: 3, 15, 75, ...

$$3, 15, 75, \dots$$

→ Divide consecutive terms: $r = a_2 / a_1 = 15 / 3 = 5$.

→ Verify: $75 / 15 = 5$. The sequence is geometric.

Answer: $r = 5$

19. A geometric sequence has first term $a_1 = 5$ and common ratio $r = 3$. Find the 4th term.

$$a_1 = 5, r = 3, n = 4$$

→ Use the formula $a_n = a_1 * r^{(n - 1)}$.

$$\rightarrow a_4 = 5 * 3^{(4 - 1)} = 5 * 3^3.$$

$$\rightarrow = 5 * 27 = 135.$$

Answer: $a_4 = 135$

20. Find the common ratio r of the geometric sequence: 1, 4, 16, ...

$$1, 4, 16, \dots$$

→ Divide consecutive terms: $r = a_2 / a_1 = 4 / 1 = 4$.

→ Verify: $16 / 4 = 4$. The sequence is geometric.

Answer: $r = 4$

21. A geometric sequence has first term $a_1 = 3$ and common ratio $r = 2$. Find the 7th term.

$$a_1 = 3, r = 2, n = 7$$

→ Use the formula $a_n = a_1 * r^{(n - 1)}$.

$$\rightarrow a_7 = 3 * 2^{(7 - 1)} = 3 * 2^6.$$

$$\rightarrow = 3 * 64 = 192.$$

Answer: $a_7 = 192$

22. Find the common ratio r of the geometric sequence: 2, 10, 50, ...

$$2, 10, 50, \dots$$

→ Divide consecutive terms: $r = a_2 / a_1 = 10 / 2 = 5$.

→ Verify: $50 / 10 = 5$. The sequence is geometric.

Answer: $r = 5$

23. A geometric sequence has first term $a_1 = 5$ and common ratio $r = 2$. Find the 6th term.

$$a_1 = 5, r = 2, n = 6$$

→ Use the formula $a_n = a_1 * r^{(n-1)}$.

$$\rightarrow a_6 = 5 * 2^{(6-1)} = 5 * 2^5.$$

$$\rightarrow = 5 * 32 = 160.$$

Answer: $a_6 = 160$

24. Find the common ratio r of the geometric sequence: 1, 4, 16, ...

$$1, 4, 16, \dots$$

→ Divide consecutive terms: $r = a_2 / a_1 = 4 / 1 = 4$.

→ Verify: $16 / 4 = 4$. The sequence is geometric.

Answer: $r = 4$

25. A geometric sequence has first term $a_1 = 3$ and common ratio $r = 3$. Find the 7th term.

$$a_1 = 3, r = 3, n = 7$$

→ Use the formula $a_n = a_1 * r^{(n-1)}$.

$$\rightarrow a_7 = 3 * 3^{(7-1)} = 3 * 3^6.$$

$$\rightarrow = 3 * 729 = 2187.$$

Answer: $a_7 = 2187$

26. Find the common ratio r of the geometric sequence: 4, 12, 36, ...

$$4, 12, 36, \dots$$

→ Divide consecutive terms: $r = a_2 / a_1 = 12 / 4 = 3$.

→ Verify: $36 / 12 = 3$. The sequence is geometric.

Answer: $r = 3$

27. A geometric sequence has first term $a_1 = 4$ and common ratio $r = 2$. Find the 7th term.

$$a_1 = 4, r = 2, n = 7$$

→ Use the formula $a_n = a_1 * r^{(n-1)}$.

$$\rightarrow a_7 = 4 * 2^{(7-1)} = 4 * 2^6.$$

$$\rightarrow = 4 * 64 = 256.$$

Answer: $a_7 = 256$

28. Find the common ratio r of the geometric sequence: 4, 16, 64, ...

$$4, 16, 64, \dots$$

→ Divide consecutive terms: $r = a_2 / a_1 = 16 / 4 = 4$.

→ Verify: $64 / 16 = 4$. The sequence is geometric.

Answer: $r = 4$

29. A geometric sequence has first term $a_1 = 4$ and common ratio $r = 4$. Find the 7th term.

$$a_1 = 4, r = 4, n = 7$$

→ Use the formula $a_n = a_1 * r^{(n - 1)}$.

$$\rightarrow a_7 = 4 * 4^{(7 - 1)} = 4 * 4^6.$$

$$\rightarrow = 4 * 4096 = 16384.$$

Answer: $a_7 = 16384$

30. Find the common ratio r of the geometric sequence: 1, 2, 4, ...

$$1, 2, 4, \dots$$

→ Divide consecutive terms: $r = a_2 / a_1 = 2 / 1 = 2$.

→ Verify: $4 / 2 = 2$. The sequence is geometric.

Answer: $r = 2$
